

# CFD 2024-25 – Lab 01

## Exercise 1

Let us consider the following diffusion-reaction problem defined in the unit square  $\Omega = [0, 1]^2$ :

$$\begin{cases} -\Delta u + 3u = 2(y^2 - 1)e^{-x}, & \text{in } \Omega, \\ \nabla u \cdot \mathbf{n} = -y^2, & \text{on } \Gamma_1 = \partial\Omega \cap \{x = 0\}, \\ \nabla u \cdot \mathbf{n} = y^2, & \text{on } \Gamma_2 = \partial\Omega \cap \{x = 1\}, \\ \nabla u \cdot \mathbf{n} = 0, & \text{on } \Gamma_3 = \partial\Omega \cap \{y = 0\}, \\ \nabla u \cdot \mathbf{n} = 1, & \text{on } \Gamma_4 = \partial\Omega \cap \{y = 1\}. \end{cases}$$

- 1 Write the weak formulation of the differential problem.
- 2 Use the formulation of the previous point in order to obtain the finite elements formulation and the corresponding algebraic problem.
- 3 Solve the problem using the variational approach in Firedrake (use the `solve` command). Use the space  $\mathbb{P}^1$  of continuous piecewise linear functions.
- 4 Extract the matrices and vectors associated to the finite element approximation of the problem and solve the problem algebraically.

## Exercise 2

Let us consider the Laplace problem defined in the unit square  $\Omega = [0, 1]^2$

$$\begin{cases} -\Delta u = 0, & \text{in } \Omega, \\ u = y(1 - y), & \text{on } \Gamma_1 = \partial\Omega \cap \{x = 0\}, \\ u = 0, & \text{on } \Gamma_2 = \partial\Omega \cap \{x = 1\}, \\ \nabla u \cdot \mathbf{n} = 1, & \text{on } \Gamma_3 = \partial\Omega \cap \{y = 0\}, \\ \nabla u \cdot \mathbf{n} = 0, & \text{on } \Gamma_4 = \partial\Omega \cap \{y = 1\}. \end{cases}$$

- 1 Write the weak formulation of the differential problem, enforcing Dirichlet boundary conditions in the strong sense.
- 2 Derive the finite element formulation of the problem.  
From now on, consider **piecewise linear** finite elements.
- 3 Implement and solve the problem in Firedrake.
- 4 Extract the matrices and vectors associated to the finite element approximation of the problem. What is the structure of the stiffness matrix?